Programming HW3 Discussion

a) **Hardware & software** used:

- **Operating System**: MacOS Seqouia 15.6.1
- **App**: *PyCharm* 2025
- **Python Version**: Python Version 3.13.5
- **Python Libraries & Packages:**
  - → **pandas** – dataset handling & analysis
  - → **openpyxl** / **xlrd** – Excel file reading & writing
  - → Sklearn.cluster-KmNEa
  - → **scikit-learn -** K-means clustering & silhouette scoring
  - → **Matplotlib** – data visualization (plots)
  - → **Scipy** – distance calculate for elbow method

(b) **Features of your program** (e.g., "Used pandas DataFrames for dataset handling, loops and conditionals for searching, matplotlib for visualization").

- Loaded **Iris dataset (iris.xlsx)** using pandas for clean data handling.
- Extracted the 4 numerical features for clustering (Sepal length, Sepal width, Petal length, Petal width).
- Ran **K-Means clustering** with different cluster counts (**k = 2, 3, 4, 5, 6**).
- Computed **silhouette coefficients** for each run to evaluate cluster separation.
- Created and saved **6 pairwise scatter plots per k value** to visualize clusters (30 total plots).
- Implemented **Elbow Method** (k = 1–10) to determine the optimal number of clusters.
- Saved all outputs (plots, CSVs, silhouette scores) into organized folders by k value.
- Printed silhouette scores and dataset info to console for report screenshots and annotation.

(c) **Problems you encountered**

- **Excel Read Errors:** Initially ran into dependency issues with reading .xlsx files (needed openpyxl installed). Fixed by ensuring the correct engine was specified in pandas.
- **Syntax Errors / Debugging:** Mistyped loops, missed parentheses, and inconsistent column indexing caused runtime errors. As expected most time spent here.
- **Plot saving conflicts:** Overwriting or missing subfolders occurred until directories were created automatically with os.makedirs().
- **Runtime confusion:** 30 scatter plots took time to generate; had to organize naming conventions carefully.
- **Silhouette calculation edge cases:** For certain k values, silhouette scoring failed when clusters had too few points — handled with try/except.

(d) **Assigned discussion problems:**

None

(e) **Fill in a reflection table**:

Item	Answers
Total time spent on Assignment	I forgot to time it lots of time but a little less than usual ones maybe 10 – 12 hrs maybe because I worked slow sporadically due to illness. Annotations take a longer time too, organizing all the data in different files.
Total Time spent on corrections	Several hours fixing indexing errors, adjusting plot layouts, and organizing folder outputs. Most spent here, fixing stuff.
Problems & difficulties:	Minor syntax issues at first, setting up the correct package versions, and keeping all the output files organized. Also making sure all 30 scatter plots were properly saved and labeled took patience.
Reflections (good & bad)	**Good**: gained hands-on experience with unsupervised learning, clustering metrics, and visualization. Improved workflow for saving and organizing outputs. **Bad**: took a lot of time to generate, save, and organize all plots and outputs. Running multiple k-values was repetitive, and I had to double-check folder outputs carefully.
Comments / Suggestions	I think this one had the clearest instructions compared to previous homework assignments. It was much easier to follow the steps. I just hope I

	haven’t missed anything.

HW Programming #3: - Data Analytics with K-Means Clustering

Tasks: Experiment with the K-means clustering technique on ‘iris.xlsx’ dataset.

Assignment Instructions:

Run the K-Means clustering technique on the sheet 1 of ‘iris.xlsx’ dataset from your local directory

(1) [40%] Run with k = 2, k= 3, k = 4, k = 5, and k = 6, respectively

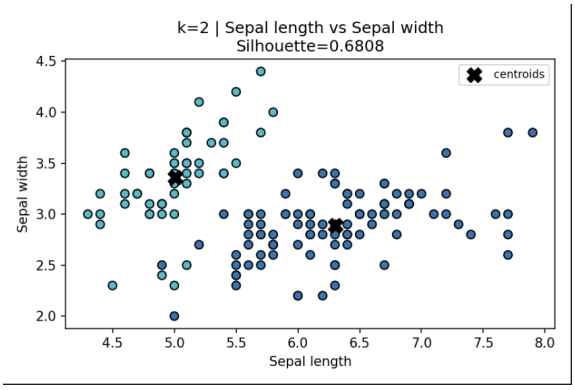
(2) [10%] Get the silhouette coefficients for each run of the k values, record the silhouette coefficients of the runs for the different k values in a table, and include the table in your Submission.

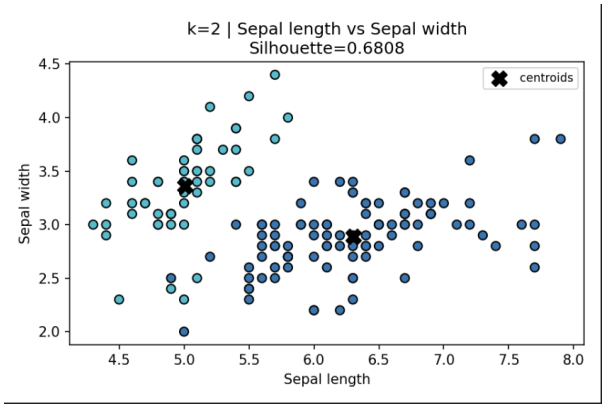
```
Run    KrishaGurung_HW3_KMeansClustering
/Users/krishagurung/PycharmProjects/PythonDataAnalytics/HW#1/.venv/bin/python /User
Loading data...... iris.xlsx
Features detected: ['Sepal length', 'Sepal width', 'Petal length', 'Petal width']
Data Shape: (150, 4)

Running KMeans with k = 2 ...
Silhouette score (k=2): 0.6808

Running KMeans with k = 3 ...
Silhouette score (k=3): 0.5526

Running KMeans with k = 4 ...
Silhouette score (k=4): 0.4978

Running KMeans with k = 5 ...
Silhouette score (k=5): 0.4885

Running KMeans with k = 6 ...
Silhouette score (k=6): 0.3682
```

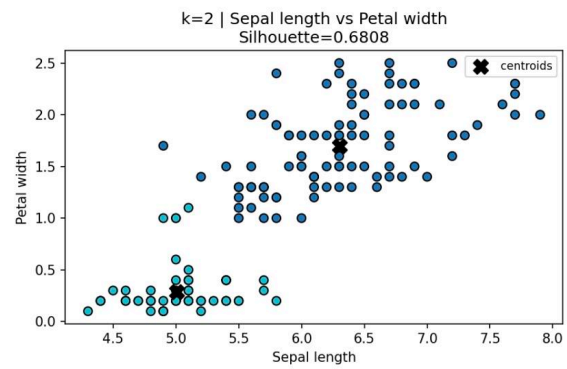
Comment: Silhouette coefficients measure cluster quality (closer to 1 = better).
As we can see above, the best clustering occurs at **k = 2**, with the score gradually decreasing as k increases.

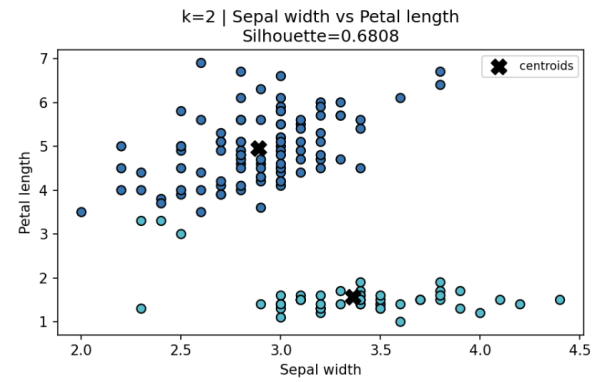
(3) [30%] Plot the clustered data in each of the attribute pairs (total 6 plots for the 6 pair combinations) for each run of the 5 k values, i.e., there should be a total of 5 x 6 = 30 plots in your submission.

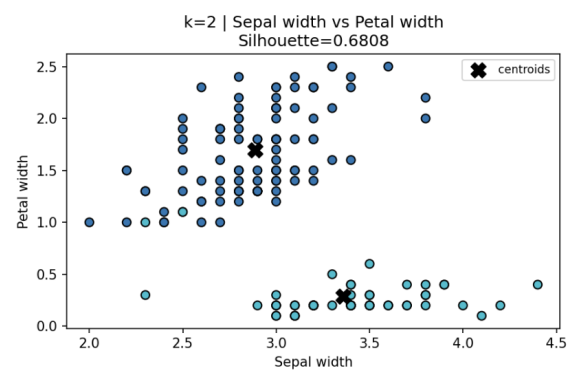
K = 2

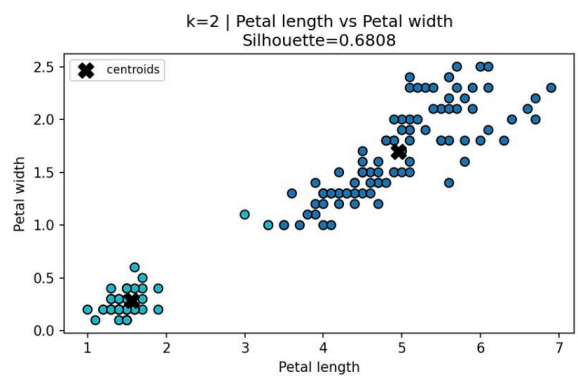






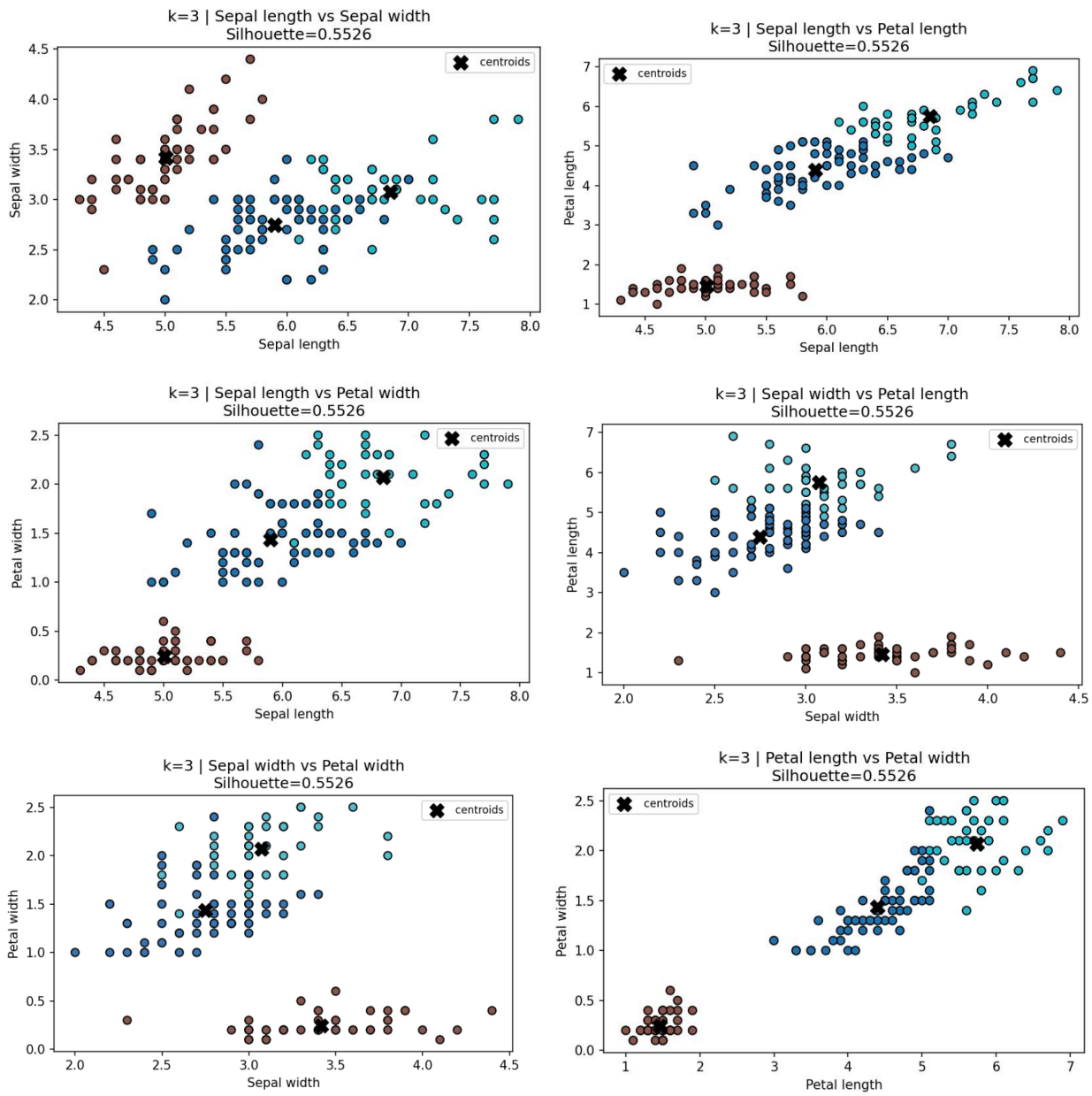






For k = 2, the dataset is divided into two broad groups, showing mostly a clear separation between flowers with smaller vs larger petals. This gives the **highest silhouette score (≈0.68)**, meaning clusters are well separated.

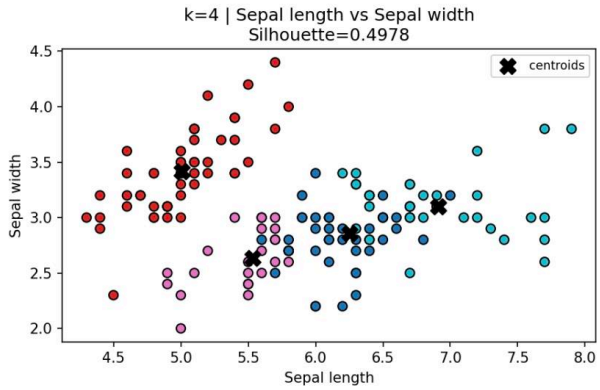
K = 3

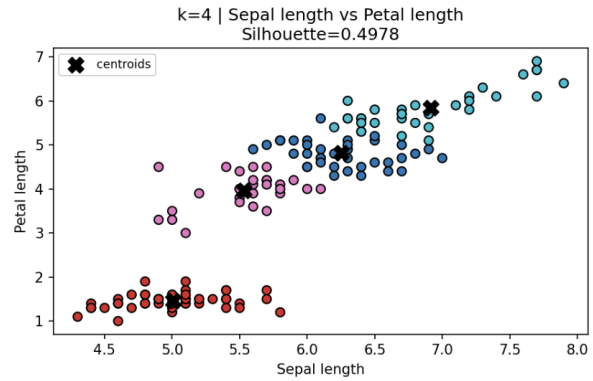


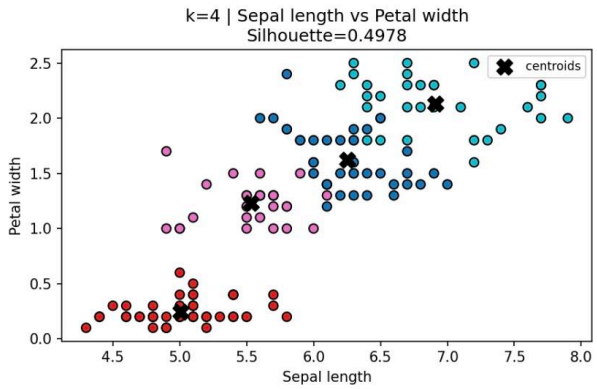
Comment: For k = 3, data separates into 3 visible groups for the most part. I'd say the clusters align closely with the three Iris species. One cluster represents *setosa*, and the other two divide *versicolor* and *virginica*. Still has a strong silhouette score (≈0.55).

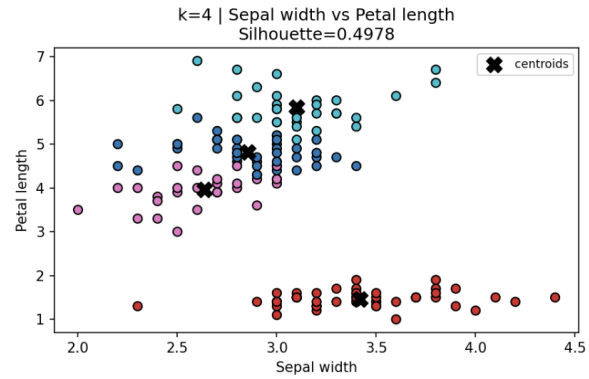
Centroids (black X's) show clear cluster centers for sepal vs. petal dimensions.
Some overlap suggests similar flower types.

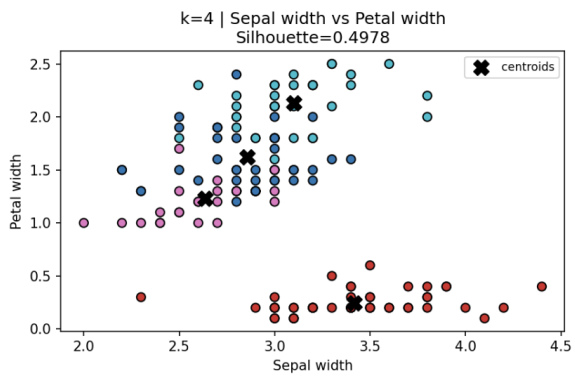
K = 4

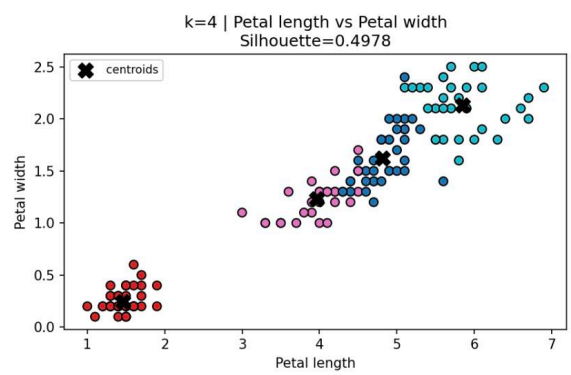






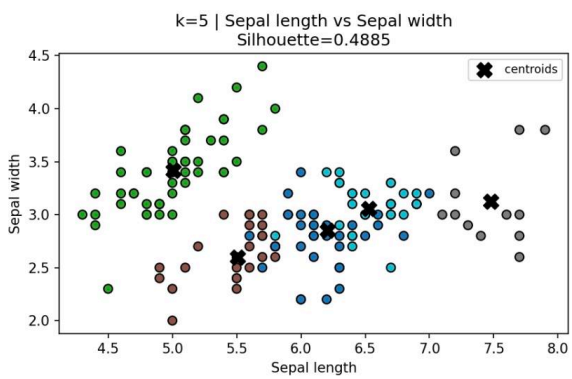


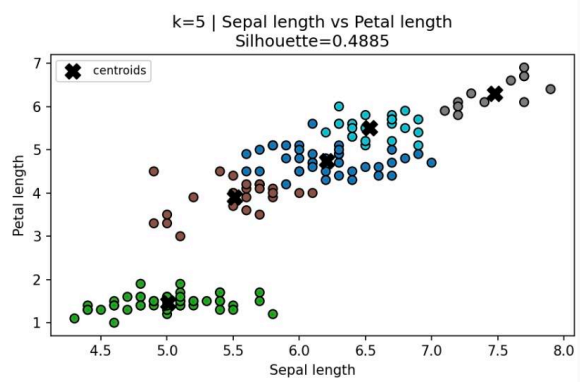


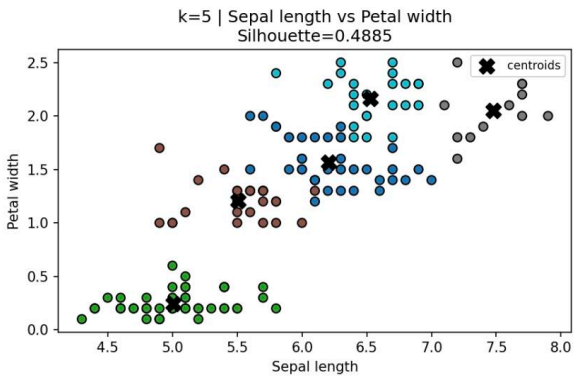


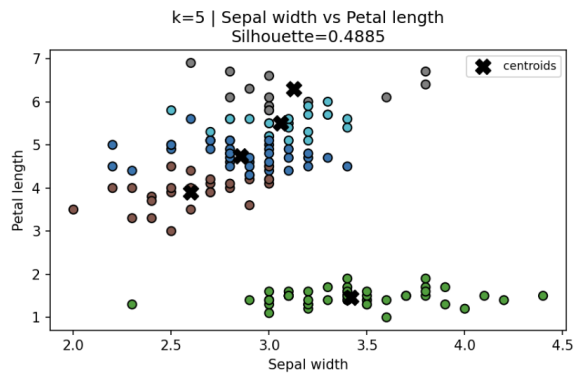
Here, with k = 4, a large cluster begins to split further. Slightly lower silhouette score (≈0.50), meaning clusters start overlapping more.

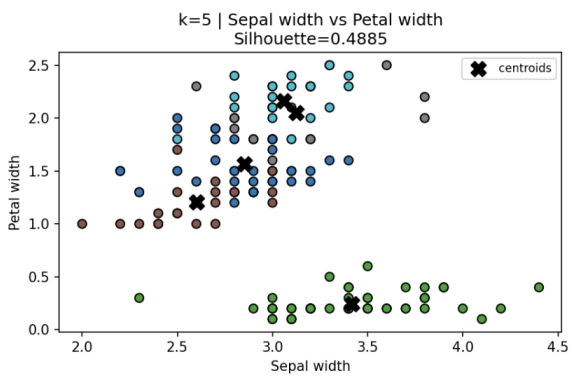
K = 5

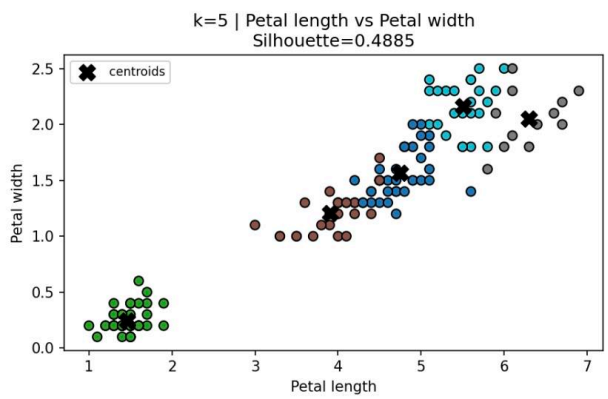






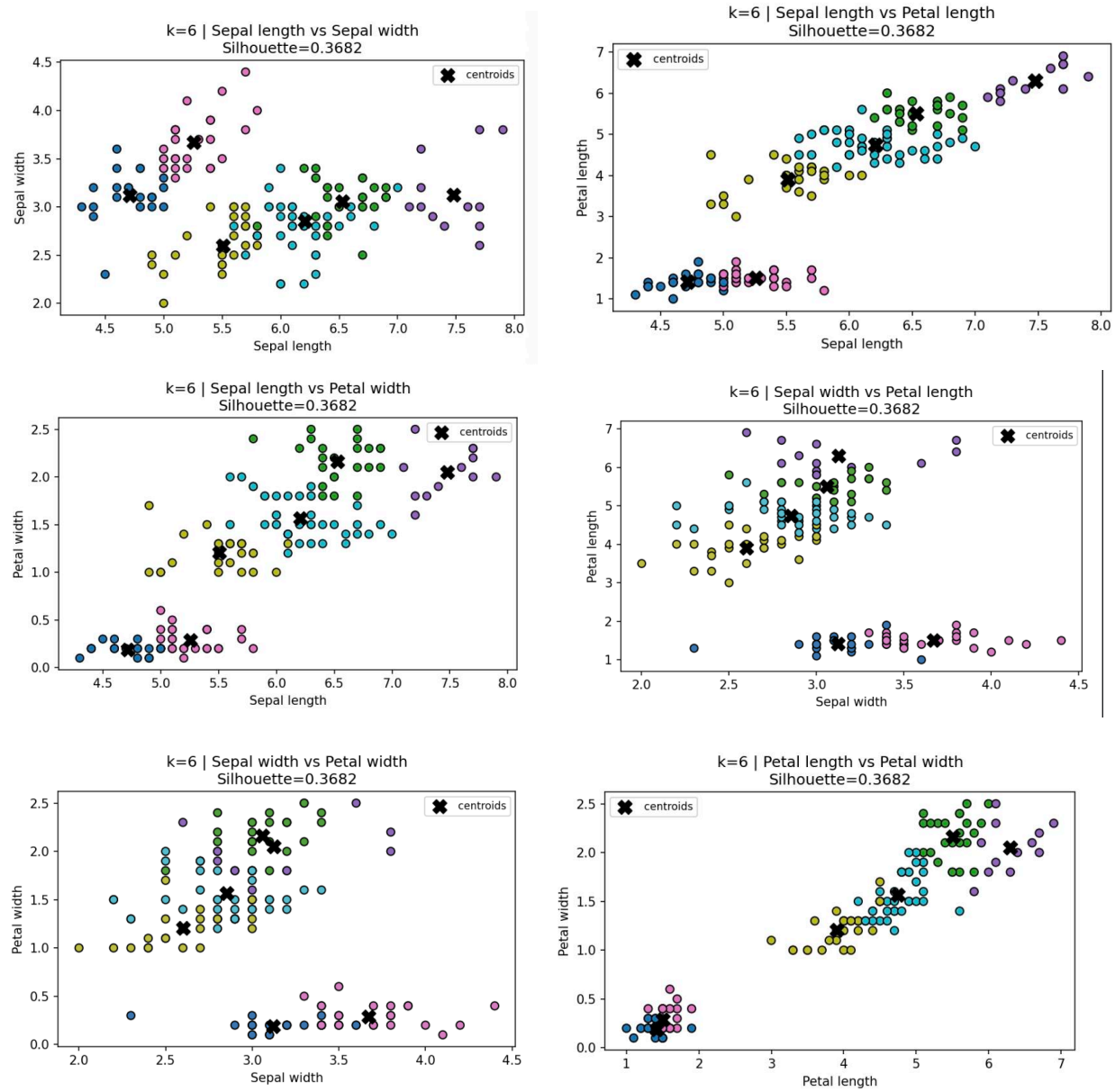






With k = 5, we see that some clusters become very small and overlapping increases. The silhouette score drops again (≈0.49), suggesting that 5 clusters might be too many for this dataset.

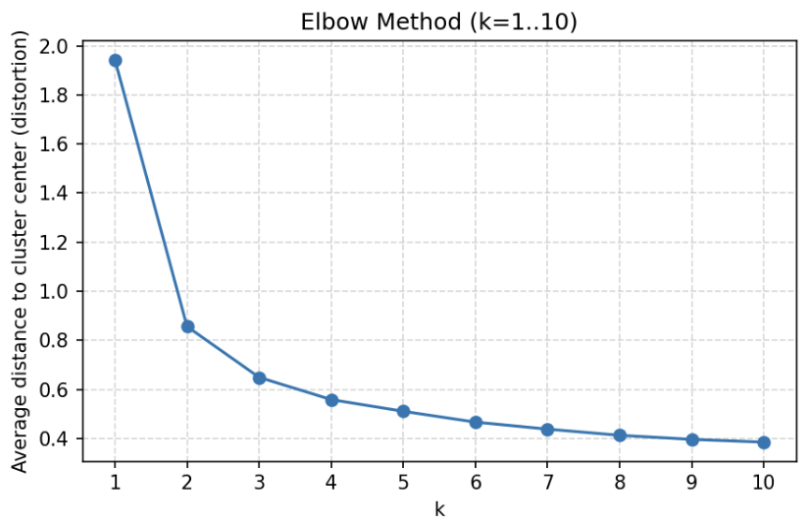
K = 6



Here, we see a lot of overfitting and overlapping. Clusters are fragmented and less meaningful. The lowest silhouette score (≈0.37) indicates very poor separation.

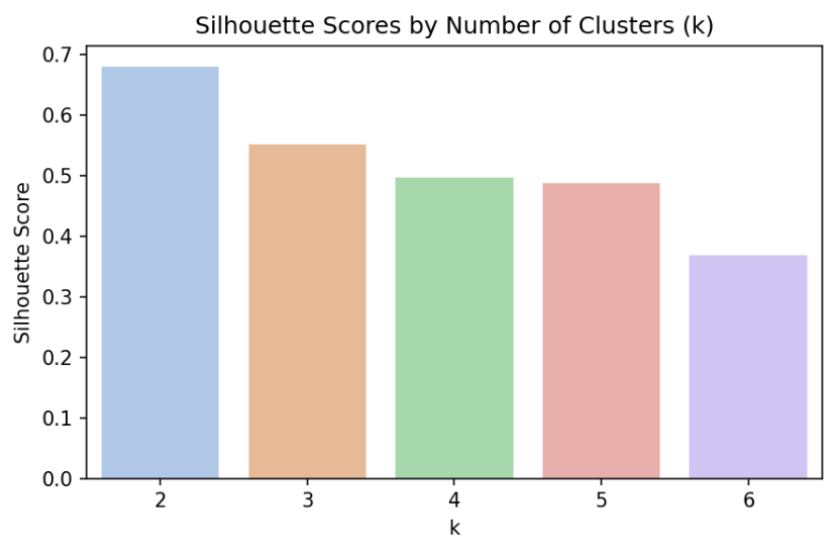
(4) [20%] Based on the test runs of your program to find the best k value for clustering on the given dataset, and compare it with the optimal k value obtained from the elbow method. Show the plot of elbow method with k values from 1 to 10 on the given dataset.

Elbow Plot:



The elbow method plot shows a noticeable "bend" near **k = 3**, suggesting that it's the optimal cluster count before diminishing returns.
However, the silhouette scores showed k=2 as the strongest. This difference can occur due to variations in how compact vs. well-separated clusters are.

Silhouette Scores:



Summary:

Based on the silhouette method, the best clustering is at **k=2**.

The elbow method suggests **k=3** as the optimal trade-off between compactness and simplicity.
Together, these results imply 2–3 distinct natural groupings in the Iris dataset — consistent with known Iris species patterns.

Python libraries needed: numpy, pandas, sklearn.cluster-KMeans, klearn.metrics-silhouette_score, matplotlib.pyplot, scipy.spatial.distance-cdist